Linear Free Energy Relationships

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Abstract

The change in the difference between the free energy of a reactant and the transition state of a reaction with changing substituents should have a correlation with similar changes in ΔG in a related system. As a result, the effect on reaction rates or equilibrium constants due to changing substituents in a reactant should have a correlation to the effect in a different reaction. Ways to measue and standardize the effect of a substituent on a reaction and ways to determine the magnitude of the effect in different reactions will be discussed in this document.

Contents

1 Introduction.

How does the magnitude of perturbing a reaction relate to the magnitude of the same perturbations of another comparable reaction?

If there is a linear relationship between the rates of a standard reaction system and the rates of the new reaction system, then the perturbations affect the transition state in the same manner for both reaction systems. The magnitude of this effect will likely be different.

Reaction systems where the transition state has more or less polar character than the initial state, and equilibria where the final state has more or less polar character than the initial state, will have the rate constant, or equilibrium constant, affected by changes in electron density.

To establish the relative effects of structural changes in a molecule on the electron density at the reaction center we will take one reaction (preferably one that is easy to follow) as a standard and make a series of structural changes to the reactant that contains the reaction center. We will state that the relative change in equilibrium constant reflects the magnitude of changes in the electron density at the reaction center.

2 Hammett Equation.

For the effect of increasing or decreasing the electron density at the reaction center we use the dissociation of benzoic acid. The energy difference between the protonated and dissociated benzoic acid will be significantly affected by changes in electron density at the carbonyl carbon due to the formation of the negative charge. So it is reasonable to say that electron density at the carbonyl carbon is related to the equilibrium constant.

We will synthesize a series of substituted benzoic acids and the relative equilibrium constant (compared to unsubstituted benzoic acid) will give us a measure of how the electron density at the carbonyl carbon is affected by each substitution. We don't really know exactly how electron drawing or donating each substituent is. We might know how electronegative a substituent is but just how electron withdrawing or donating it is for affecting electron density at the reaction center in a particular reaction will be related to how that substituent affects the equilibrium for benzoic acid.

The log of the ratio of the equilibrium constant for the dissociation of a substituted benzoic acid to the equilibrium constant for the dissociation of the unsubstituted benzoic acid will be the measure for how much that particular substituent affects the electron density at the reaction center. The value that describes this effect is σ.

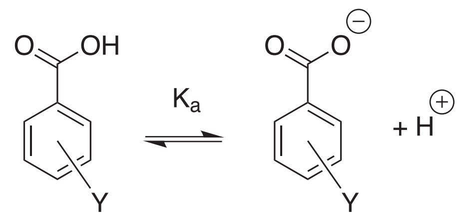


Fig. 1: Acid Equilibrium of Benzoic Acid

So for measuring the effect an electron withdrawing or donating substituent will have on the electron density at the reaction center we use the standard reaction of benzoic acid equilibrium to define the σ value for each substituent.

Using the dissociation of each substituted benzoic acid (K_Y) compared to the unsubstituted (K_0) we define σ as shown in Equation 1.

$$\sigma = \frac{K_Y}{K_0} \tag{1}$$

Tab. 1: Acid dissociation of substituted benzoic acids

Substituent (Y)	$K_a\left(\times 10^5\right)$	$\sigma = \log\left({}^{K_Y}/_{K_0}\right)$
—NO_2	35.5	0.75
—CN	27.5	0.64
—Cl	10	0.2
—H	6.3	0
—Me	4.3	–0.17
—OMe	3.4	–0.27

Now we have values for the substituent constant, σ, which describes the electron withdrawing effects of these substituents on the standard reaction system (all the values above were for p-substituted rings, m-substituents may have different σ values).

Now lets see how these substituent constants relate to another reaction system. Lets take the hydrolysis of ethyl benzoate. The transition state will have a more negative charge than the ground state ethyl benzoate so we would expect that changes in electron density at the reaction center (the carbonyl carbon) would have an effect on the rate constant for the reaction.

The rate-determining step is the k_1 step so electronic effects on the transition state for this step will be observed in changes in the value of k_1. Since the k_1 step is the rate determining step and assuming that $k_2 > k_{-1}$ we can state the simple rate expression shown in Equation 2. So how does k_1 change as we use

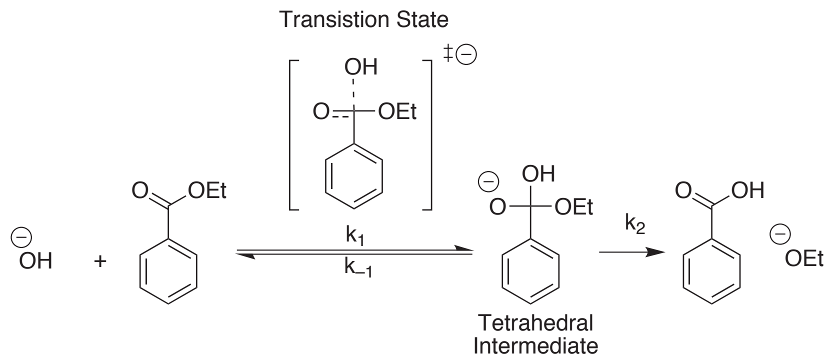


Fig. 2: Mechanism of Ethyl Benzoate Basic Hydrolysis in Water

various substitutions on the phenyl ring? Let us plot the log of the observed rate constant for hydrolysis (Table 2) against the substituent constant, σ, for each substituent (see Figure 3 on the next page.)

$$\frac{d\left[P\right]}{dt} = k_1\left[B_zOEt\right]\left[OH\right] \tag{2}$$

Tab. 2: Rate of Hydrolysis of a Series of Substituted Ethyl Benzoates

Substituent (Y)	k_{obs} $\left(M^{-1}s^{-1}\right)$
—NO_2	32.9
—CN	15.7
—Cl	2.10
—H	0.289
—Me	0.172
—OMe	0.143

We have a linear correlation so we can see that the electronic effects on the acid equilibrium of benzoic acid for each substituent correlate with the electronic effects of each substituent on the rate of hydrolysis for ethyl benzoate. Notice that the slope of the line is 2.41. This indicates that the substituent effect has a greater importance for this system than the reference system. If the slope were 1.0, we would know that the effect of each substituent was exactly the same (on average) for the observed reaction as compared to the reference reaction.

The correlation constant for this system compared to the reference system is $\rho = 2.41$. This is the reaction constant, ρ.

In systems that have a linear correlation with another reference system we have a linear free energy relationship (the change in free energy of the transition state

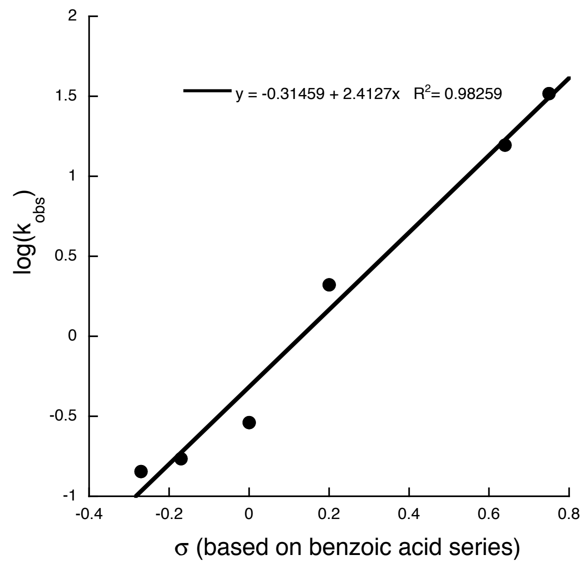


Fig. 3: Hammet plot for base hydrolysis of ethyl benzoate.

or equilibrium reaction due to a change in structure is directly proportional to the change that that structural change will have in another, related system.)

Over time scientists have averaged the σ values for the possible substituents in many, many different systems and come up with a standardized set of σ values for various substituents. These values are very similar to the Hammett σ values derived from the acid dissociation of benzoic acid but give slightly better correlations in other related systems. A Hammett plot of ethyl benzoate hydrolysis using the modern σ values is shown in Figure 4 on the following page.

Obviously there was not a tremendous improvement in this case but it is standard practice to use the newer "tweaked" σ values rather that Hammett's original values. It will improve correlations in most systems. See Table 3 on page 7 for a comparison of Hammett σ and modern σ values.

This linear relationship can be expressed for systems containing substituted benzene rings that are close enough to the reaction center to have an effect on its electron density (like the acid equilibria of substituted benzoic acids). This relationship is the Hammett equation.

$$\log\left(\frac{K_Y}{K_0}\right) \; or \; \log\left(\frac{k_Y}{k_0}\right) = \rho \cdot \sigma \tag{3}$$

The log of the relative rate constant (or equilibrium constant) is directly proportional to the σ value for the particular substituent. The proportionality

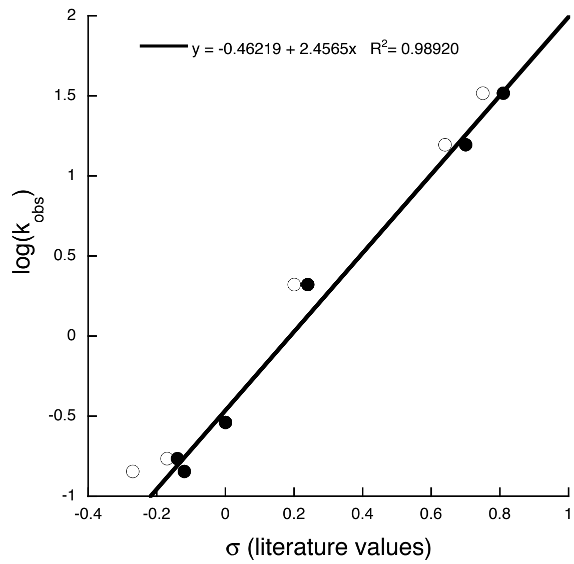


Fig. 4: Hammett plot for base hydrolysis of ethyl benzoate using standard σ values (closed circles, $\bullet$). The Hammett σ values from Table 1 on page 3 that were used in Figure 3 are shown as open circles ($\circ$) for comparison.

constant ρ, the reaction constant, gives the actual relationship of the reaction to the reference reaction (see Equation 3).

Note that Figure 4 plots the log of the rate constant, k_Y (k_{obs} for each substituent, Y), versus σ. This is the usual way of plotting a free energy relationship in modern publications. We can use the line equation (the slope is ρ and the Y-intercept is $\log(k_0)$) to find the likely rate constant for a substituted ethyl benzoate we have not yet tested if we know the σ value for that substituent from another source (a table of σ values for example). Using the relative rate constant (k_Y/k_0) would not alter the slope, ρ, of the plot but would force the Y intercept to pass through the origin (because $k_0/k_0 = 1$ and $\log(1) = 0$).

Using the line equation from Figure 4 which is...

$$\log(k_Y) = \log(k_0) + \rho \cdot \sigma \tag{4}$$

You can see how this is easily rearranged to the classic Hammett relative rate equation...

$$\log\left(\frac{k_Y}{k_0}\right) = \rho \cdot \sigma \tag{5}$$

The Y intercept for Equation 4 is $\log(k_0)$. The Y intercept for Equation 5 is 0 (the origin). the slope in both equation is ρ. So you can see that it makes no

Tab. 3: Comparing Hammet σ values (from Table 1 on page 3) with modern σ values.

Substituent (Y)	σ (Modern)	σ (Hammett)
—NO_2	0.81	0.75
—CN	0.70	0.64
—Cl	0.24	0.2
—H	0	0
—Me	–0.14	–0.17
—OMe	–0.12	–0.27

difference whatsoever if you plot the log of the rate constant vs. σ or the log of the relative rate constant vs. σ. Compare the two in Figure

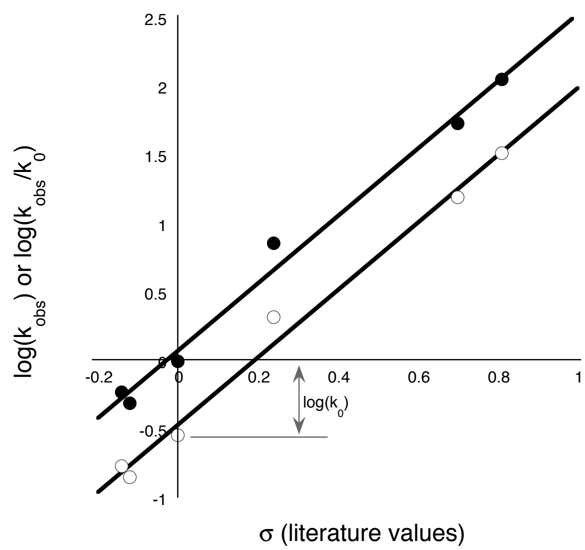


Fig. 5: Data from Figure 4 expressed as relative (•) and absolute (○) rate constants

The graph is basically the same; it is just moved vertically up the Y-axis by a value equal to $\log(k_0)$. (Note: The point for the unsubstituted ethyl benzoate falls exactly on the origin here. But the line does not pass through the origin (Y intercept is 0.18). This is due to the curve fit algorithm finding the best line through the slightly scattered points).

3 Changing Reaction Mechanisms

If the changes to the rate of the rate determining step due to changing substituents leads to that step no longer being the rate determining step then we may see deviations from the straight line. The rate of a simple multistep reaction depends on the rate of the slowest step. If the rate determining step changes then we are looking to another transition state that is involved in the new rate determining step (RDS). Will the transition state (TS) of the new RDS have exactly the same dependance of changing ring substituents as the TS of the original RDS? Unlikely, so the slope of the line, ρ, would likely change when the RDS changes.

Also, as we change the electronic density at the reaction center we are changing the energy of the TS. We may be raising the free energy of the TS as we increase the electron withdrawing character of the reactions (thus slowing the reaction) until that TS is higher in energy than the TS of another possible reaction mechanism. The reaction would proceed through the new TS, which is now the lowest TS. Thus the mechanism will change.

So we have two explainations for a change in the slope of a Hammett plot:

Case 1: We are changing the RDS in a multistep reaction (the mechanism does not change, just the rates of each step change until we have a new RDS)

Case 2: We are changing the energy of the TS until a new TS become the easiest route to product, thus changing the mechanism. The old and new mechanisms might both be one-step reactions but they proceed through a different TS.

3.1 Case 1

We have a reaction and we are accelerating the RDS as we use substituents with larger and larger σ values. So the slope (ρ) of the Hammett plot in the region controlled by the original RDS is positive. Then the step becomes so fast that it is no longer the RDS step. Another, slower step now controls the rate of product formation (we are assuming a simple multistep reaction where all reactions are unidirectional). This slower step will be affected by changes in electron density to a lesser degree than the first RDS. (if it was affected to the same or greater degree than the RDS step would never change. Its rate has not increased as much as the rate of the original RDS has increased, which is why it is the new RDS.) So the new slope (ρ) in the region controlled by the second new RDS will be lower the first slope (ρ). ρ might even be negative if its TS was raised in energy by decreasing electron density. As a result we would expect a Hammett plot to start with a given slope and then the slope would change to a lower (or even a negative) value. The plot would be concave, opening toward the bottom of the graph.

In the acid hydrolysis of an imine, the mechanism is a two-step process. First the amino group is protonated resulting in a positively charged carbon on the imino carbon (going through a TS where the positive charge is forming on that carbon). Then water can attack the carbocation to give the hydrolysis products (going through a TS where the positive charge is disappearing from the carbon). Step 2 is the RDS in unsubstituted phenylimine systems. Obviously step 2 is accelerated by electron withdrawing substituents, which would destabilize the carbocation and decrease its energy difference relative to the TS for that step. Step 1 is the faster step in the unsubstituted system but as electron withdrawing groups are added it is decelerated because lower electron density destabilizes the forming positive charge in the TS for step 1, raising the energy of the TS and making that step slower and slower. As we use substituents with larger σ values we will accelerate the reaction while step 2 is the RDS. Eventually the rate of step 2 is increased enough and the rate of step 1 is diminished enough that step 1 becomes the RDS and it will get slower with substituents of larger and larger σ values.

Thus the Hammett plot will have a positive reaction constant, ρ, while step 2 is the slowest step and a negative ρ when step 1 is the slowest step of the two-step mechanism.

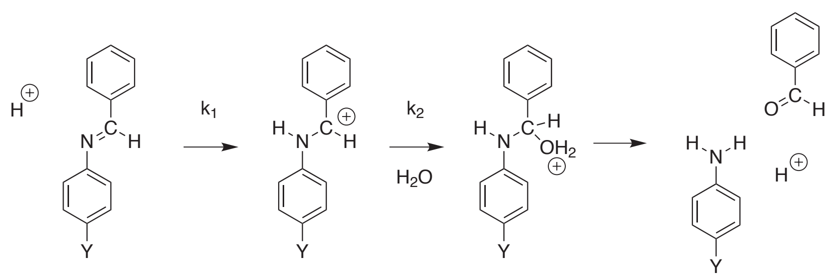


Fig. 6: Reaction Mechanism for Acid Hydrolysis of Phenyl Imines.

3.2 Case 2

We have a reaction and we are raising the energy of the transition state (TS) as we use more electron withdrawing substituents (larger σ values). As we raise the energy of the TS we obviously are slowing the reaction so we would have a negative slope (ρ_1 is negative). Perhaps there is another TS that is being lowered in energy with decreasing electron density. Eventually, when the electron density has been reduced to a given point, it becomes the easiest route to the product and becomes the TS that control the rate constant for that step. Since it is lowered by substituents with larger σ values the reaction will be accelerated by substituents that are more electron withdrawing and the slope in this region of the plot, ρ_2, would be positive. Thus we would expect a Hammett

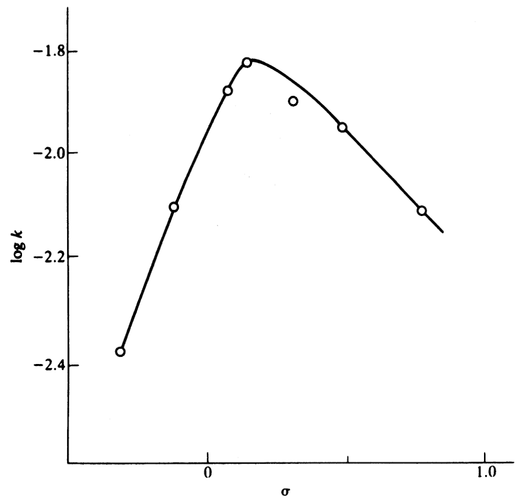


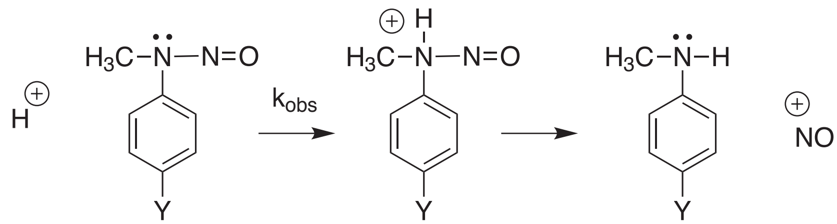
Fig. 7: Hammett Plot for Acid Hydrolysis of Phenyl Imines. (Isaacs Fig. 4.6)

plot of a system that has a changing transition state to be concave, opening toward the bottom of the plot.

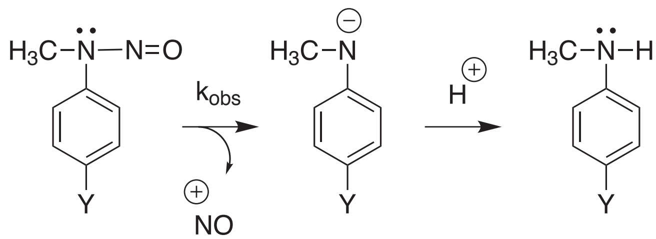
In the acid cleavage of N-nitro aniline we see a change in the reaction mechanism. There are two possible reaction mechanisms in this scheme as shown in Figure 8 on the following page. Mechanism A involves a protonation of the amino group giving a positively charged intermediate (and a positive charge forming on the amino group in the TS on the way to this intermediate) as the RDS in a two step process that leads to the product (the second step is very fast). This mechanism is an electrophilic bimolecular substitution, and SE2 mechanism (like SN2 but with electron moving toward the incoming group) Mechanism B involves the nitro group leaving on its own giving a negatively charged intermediate (and thus a negatively charge forming on the amino group in the TS). The second fast step is the protonation of the amino anion (very, very fast in water). The RDS for this mechanism is a unimolecular electrophilic substitution (like an SN1 reaction but the incoming group accepts electrons from the reaction center).

Using the ρ values for each region of the plot in Figure 9 on page 12 and σ values from the literature, one could determine reaction rates for substituted nitroanilines that one had not yet synthesized.

Inspection of the two plots in Figures 9 and 7 allow us to conclude that the reaction mechanisms are both two step (at least) processes where changing electronic density in the system results in a change in the rate determining step in the first case and that the reaction mechanism is changing and a new reaction route is now leading to the product when electron density has been changed enough.



(a) Associative mechanism for acid clevage



(b) Dissociative mechanism for acid cleavage

Fig. 8: Two Possible Mechanisms for Acid Cleavage of Nitroaniline

4 Resonance Substituent Parameters

Often one may apply the Hammett s substituent constants to a reaction system and find that some correlate well and others result in points that lie far off the line. In the solvolysis of triphenylmethyl chloride (which proceeds via and SN1 mechanism with a carbocation intermediate) we find that σ values for meta-substituents on the ring correlate fairly well with the reaction rate of those substituted triphenylmethyl chloride. But some para-substituents have a far greater effect on the rate of reaction than the Hammett σ values would indicate and thus lie above the trend. Electron donating groups stabilize the carbocation intermediate but some of the groups seem to be “super” electron donating.

It turns out that groups that can directly donate electron pairs into the resonance system will have a much greater effect than groups that donate or withdraw electrons by induction alone. Groups with a free electron pair or a conjugated double bond system will need new parameters to take into account the resonance interactions, which are far greater in magnitude than induction effects. For resonance effects to be observed the reaction center must be able to directly exchange electrons with the donator via the conjugate network. Carbocations attached to a phenyl ring can interact with substituents via resonance in the phenyl ring.

So in reactions where resonance interaction with the reaction center is possible we will need new substituent constants. Reactions that interact with resonance

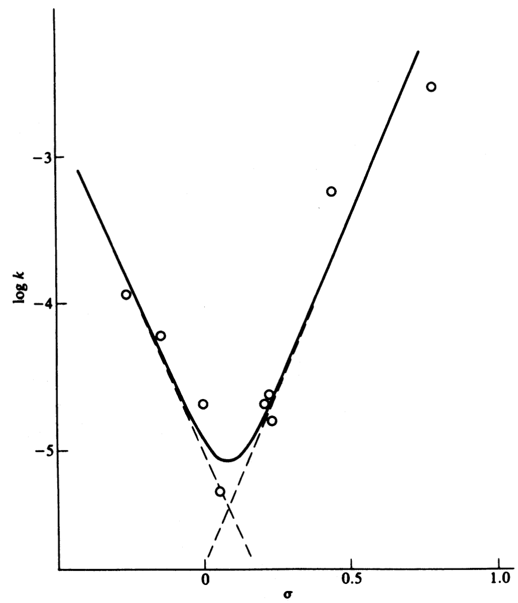


Fig. 9: Hammett Plot for Acid Cleavage of Nitroanilines (Isaacs Fig. 4.5)

electron acceptors will use one set of these new substituent constants and reactions that interact with resonance electron donors will use a different set of these new substituent constants. Reactions that interact with resonance electron donors correlate with $\sigma+$ substituent constants and reactions that interact with resonance electron acceptors correlate with σ^- substituent constants.

σ^+ and σ^- reaction constants only apply to para-substituents. *meta*-Substituents will not have a resonance connection and regular Hammett σ values will generally correlate. The σ^+ and σ^- substituent parameters were determined by using the solvolysis of cumyl chloride in acetone as a reference reaction. Measuring the reaction rates with m-substituted reactants and constructing a Hammett plot gave a reaction constant ρ = -4.54. Then σ^+ and σ^- substituent parameters were assigned to *para*-substituents so that the reaction rates for the corresponding p-substituted reactants fit on the line. These new σ^+ and σ^- give good correlations when used with appropriate reactions.

In the solvolysis of triphenylmethyl chloride, the carbocation will interact via the resonance network with resonance electron donors so we will us the σ^+ constants. The ρ value we obtain for a reaction that involves interaction with resonance electron donors is called ρ^+. Using the σ^+ constants from Table 4.1 in Isaacs, we can now replot the Hammett plot for the solvolysis of triphenylmethyl chloride. This is now a Brown-Okamoto plot (using σ^+ and σ^-).

ρ^+ values are usually negative because the more electron withdrawing (higher σ) a substituent is the more it will destabilize the carbocation intermediate.

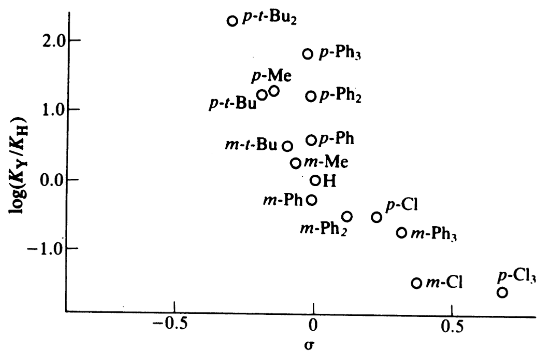


Fig. 10: Hammett Plot of Solvolysis of Triphenylmethyl Chloride (Isaacs Fig. 4.7)

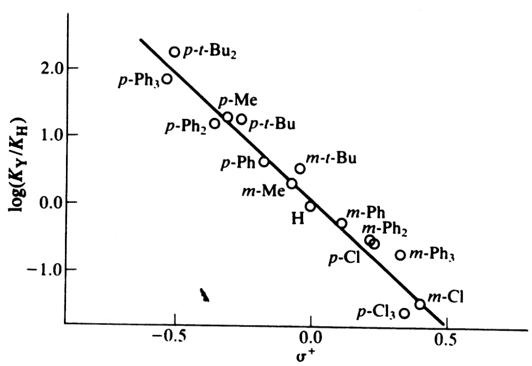


Fig. 11: Brown-Okamoto Plot of Solvolysis of Triphenylmethyl Chloride (Isaacs Fig. 4.7)

If a ρ^+ value is positive (reaction is faster with increasing σ of substituents) it is generally because the initial reactant is more positively charged than the transition state so electron withdrawing groups destabilize the reactant and raise its energy towards that of the transition state resulting in a lower energy of activation.

ρ^- values are usually positive for similar reasons.

5 Multiple Parameter Approaches

We have seen where σ is not suitable for describing the effect substitutions on a benzene ring have on the electron density of a reaction center if the reaction center is in direct resonance with the substituent. For substituents that are in direct resonance with the reaction center we use σ^+ (when the reaction center is developing an increasing positive charge. e.g. a carbocation is forming) or σ– (when the reaction center is developing a more negative charge. e.g. in the acid dissociation of aniline). When we are using the σ^+ or σ^- parameters we denote the ρ value as ρ^+ or ρ^- respectively.

It has always been a goal of physical organic chemists to describe as many reaction systems that are effected by structural changes as possible with a single method or equation. Many, many approaches have been proposed. We will discuss a few here.

5.1 Yukawa–Tsuno double reaction constant approach.

Some substituents in some reaction have a much greater dependance on resonance contribution from the substituents that might be expected from the σ^+ or σ^- parameters developed from the dissociation of cumyl chloride and the dissociation of aniline, respectively. Rather than propose yet another set of substituent parameters it has been proposed that we use a second reaction constant that would express a reaction's sensitivity to the resonance effects of substituents that have been assigned σ^+ or σ^- parameters.

This new reaction constant is denoted as r. r represents the reaction center's sensitivity to direct resonance donation. This is actually a measurement of the extent to which an empty p orbital that can overlap with the resonance system is developing in the transition state. The Yukawa-Tsuno equation is…

$$\log\left(\frac{k_Y}{k_0}\right) = \rho\left(\sigma + r\left(\sigma^+ - \sigma\right)\right) \tag{6}$$

Inspect this equation. Clearly, if $r = 0$ (the reaction center does not develop an empty p orbital in resonance with the substituent at all and is no more or less sensitive to resonance electron donation or acceptance by the substituent than the dissociation of benzoic acid) then the Yukawa-Tsuno equation will become the ordinary Hammett equation.

$$\begin{aligned} \log\left(\frac{k_Y}{k_0}\right) &= \rho\left(\sigma + 0\cdot\left(\sigma^+ - \sigma\right)\right) \qquad (7) \\ \log\left(\frac{k_Y}{k_0}\right) &= \rho\cdot\sigma \end{aligned}$$

If $r = 1$ then the reaction is equally as sensitive to resonance effects by the substituent as the reference reactions for which the σ^+ or σ^- parameters (whichever was used) were developed and the Yukawa-Tsuno equation will become the Brown-Okamoto equation.

$$\begin{aligned} \log\left(\frac{k_Y}{k_0}\right) &= \rho\left(\sigma + 1\cdot\left(\sigma^+ - \sigma\right)\right) \qquad (8) \\ \log\left(\frac{k_Y}{k_0}\right) &= \rho\cdot\sigma^+ \end{aligned}$$

Most reactions will not have exactly the same dependance on resonance contributions by *para*-substituents as the reference reactions so most reactions will have a better correlation of we fit the data to the equation using a computer to vary ρ and r until we get the best fit.

Lets examine some data selected from Table 4.5 of Isaacs. Reactions with low r values will fit the classic Hammett equation fairly well (but fit better yet when we use the Yukawa-Tsuno equation and its r term). Reactions with r values near 1 will fit the Brown-Okamoto equation quite well (but fit better yet when we use the Yukawa-Tsuno equation and its r term). Reactions with r values significantly different than 0 or 1 will not give good correlations with either the Hammett equation or the Brown-Okamoto equation as their sensitivity to the presence of a resonance donating substituent is "between" the two cases which defined each approach (dissociation of benzoic acid, ($\rho = 1.00$, $r = 0$) and dissociation of cumyl chloride ($\rho = -4.52$, $r = 1$).

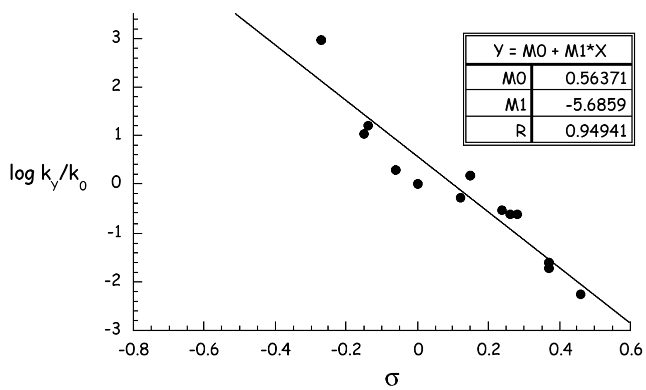


Fig. 12: Hammett Plot for Hydration of Phenyl Acetylene. $r = 0$

Tab. 4: Some Yukawa-Tsuno Reaction Constants from Table 4.5 Isaacs

Substrate	Reaction	ρ	r
Benzoic Acid	Acid Dissociation	1.00	0
Methyl Benzoate	Acid Hydrolysis	0.11	0
Cumyl Chloride ($PhCMe_2Cl$)	Hydrolysis in 90% Acetone	–4.42	1.00
Ph_2CHCl	Methanolysis	–4.02	1.23
PhCCH	Acid Hydration	–4.3	0.81
PhMeCN=NOH	Beckmann Rearrangement	–1.98	0.43

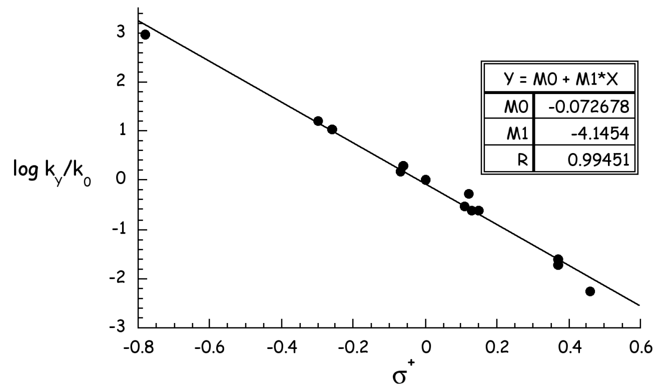


Fig. 13: Brown-Okamoto Plot for Hydration of Phenyl Acetylene. $r = 1$

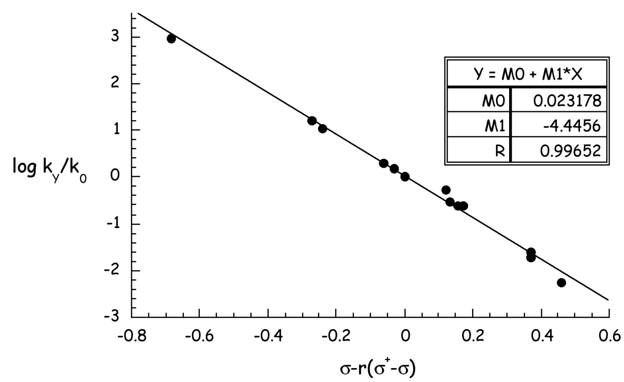


Fig. 14: Yukawa-Tsuno Plot for Hydration of Phenyl Acetylene. $r = 0.81$

6 The Taft-Ingold Approach for Aliphatic Systems

For the base hydrolysis of esters we see good correlation between a reference system (substituted benzoic acid equilibria) and other related reaction systems (hydrolysis of substituted benzoic acid esters). This is the Hammett situation.

But what is we wish to relate a greater variety of esters with a refernce acid equilibrium. Lets see if there is a LFER for the acid equilibrium of non-aromatic carboxylic acids. The data collected from one such experiment is displayed in Figure 15 on the following page

Note the very poor correlation. The Hammett approach compared systems that only differed in the electron density when various substituents were added. Because the substituents were held in the meta or para positions they did not have any effect due to steric influences. Hammett made his name by using a

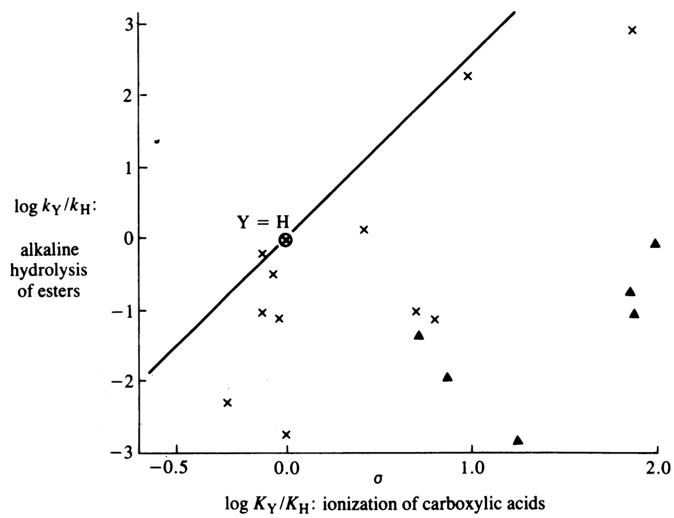


Fig. 15: Hammett-type plot of rates of hydrolysis for methyl esters of aliphatic acids and ortho-substituted benzoates vs. the relative K_a values of the corresponding carboxylic acids. Aliphatic acids, $\times$; ortho-benzoates, ▲. The slope, ρ, is 2.55. (Figure 4.10, Isaacs)

case that was guaranteed to have linear relationships due to substituent effects. Now we will begin looking at systems that are influenced by both electronic and steric effects. We will need at least two parameters, one for electronic effects due to substitution and one for steric effects due to substitution. Electron effects will be the same as in the case of the Hammett approach but steric hindrances will generally slow down the reaction. Thus the acid ionization equilibria of most carboxylic acids (ionization does not depend significantly on steric effects; a proton is very small) do not correlate with the hydrolysis of esters of those carboxylic acids (nucleophiles are much, much larger than protons). The hydrolysis reactions tend to be slower than would be expected (and thus are on the right hand side of the line in the graph in Figure 15 which is the line that would be expected if electronic effects alone ruled the relationship $\rho = 2.55$, see Fig. 4.2 in Isaacs).

To find a way to correlate structural changes in a reaction with systems that involve steric effects of the substitutents, Taft proposed two new substituent parameters:

σ^*– the electronic effect of the substituent.

E_s – the steric effect of a substituent.

These new parameters will be used in correlations with the Taft-Ingold equation...

$$\log \frac{k_Y}{k_0} = \rho^* \cdot \sigma^* + \delta \cdot E_s \tag{9}$$

where ρ^* is the reaction constant that defines the sensitivity of the reaction to the electronic effects of the substituents and δ is the reaction constant that defines the sensitivity of the reaction to the steric effects of the substituents.

First we must establish a set of substituent constants, σ* and Es, for some standard reaction. Taft used the acid and base hydrolyses of ethyl benzoate to establish these values for a series of substituents.

6.1 Taft Steric Substituent Parameters

First we look to the acid hydrolysis mechanism...

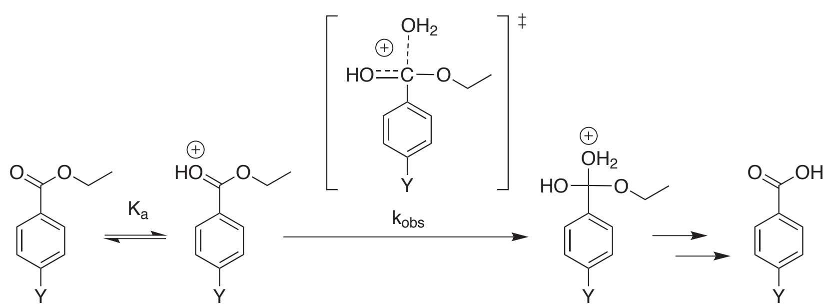


Fig. 16: Acid hydrolysis of an ester

Substitutions of the benzene ring have little effect in the acid hydrolysis of benzoates. We can see from table 4.3 in Isaacs that the reaction constant for acid ester hydrolysis is low ($\rho = 0.11$) indicating little influence on the reaction rate due to electronic effects of substituents on the benzene ring. So changes in electron density at the reaction center have little effect on the reaction rate. Thus any effects of substituents on the carbonyl carbon, whether aliphatic groups or benzene rings, will be mostly due to steric effects. We can use the acid hydrolysis of various aliphatic esters to establish the steric substituent constants for ester hydrolysis.

Why is the reaction constant for ester hydrolysis of substituted benzoic acids so low? The rate equation for the acid hydrolysis scheme above is ${}^{dP}/_{dt} = K_a \cdot [H] \cdot [ester]$ (acid pre-equilibrium combined with the rate-determining nucleophilic attack by water). Decreasing electronic density would increase K_a (disfavour the protonation. Remember, K_a is the K_{eq} for dissociation of the protonated carbonyl group) and reduce the concentration of the activated, protonated form of the ester. Of course, decreasing electron density will accelerate the nucleophilic attack of water. Each step shares the same intermediate and pass through similar transitions states so the magnitude of the electronic effect is roughly the same for both. Thus the electronic effect cancels itself out by having opposite effects of similar magnitude on the two steps of the mechanism.

So we measure the rates of acid hydrolysis for various esters using ethyl acetate (R = —CH_3) as the standard reaction rate (k_0).

$$\log \frac{k_Y}{k_0} = E_s \tag{10}$$

6.2 Taft Electronic Substituent Constants

We now have a set of steric substituent constants. We now can establish σ^* for each R group on the carbon center by using rates from base hydrolysis. For the standard reaction system of basic ester hydrolysis we set the steric reaction constant, δ, to 1. We will set the electronic substituent constant, $\rho^* = 2.54$, which is the ρ for Hammett benzoic ester hydrolysis, so that Taft ρ^* values will be comparable to Hammett ρ values. (Obviously, for acid hydrolysis we had set $\rho^* = 0$ as we were treating the system as if there was no magnitude to the electronic effect.) So we know $\log\left({}^{k_Y}/_{k_0}\right)$ (from our experiments), E_s (which we established from acid hydrolysis), ρ^* and δ (defined as $\rho^* = 2.54$ and $\delta = 1$, respectively, for ester hydrolysis) and we will be able to use the Taft-Ingold equation to set σ^*. Let us substitute our refernce values for ρ^* and δ into Equation 9 on the previous page.

$$\begin{aligned} \log \frac{k_Y}{k_0} &= \rho^* \cdot \sigma^* + \delta \cdot E_s \\ \log \frac{k_Y}{k_0} &= 2.54 \cdot \sigma^* + 1 \cdot E_s \end{aligned} \tag{11}$$

We can substitute Equation 10 into Equation 11 to get Equation 12. Using this equation we can determine σ^* from a set of base hydrolysis data and a set of corresponding acid hydrolysis data.

$$\left(\log \frac{k_Y}{k_0}\right)_{base} = 2.54 \cdot \sigma^* + \left(\log \frac{k_Y}{k_0}\right)_{acid} \tag{12}$$

Consider the mechanism of the base hydrolysis reaction in Figure 17 on the following page. We can see that the transition state for the rate determining step is almost exactly the same as the TS for the RDS of acid hydrolysis.

The structure of the transition states differ only in two little protons. So it is reasonable to state that steric effects will be very similar for both. Thus we can use the steric substituent constants established from acid hydrolysis (where we ignored the small changes due to electronic effects of substituents) and use those E_s values to establish the σ^* value.

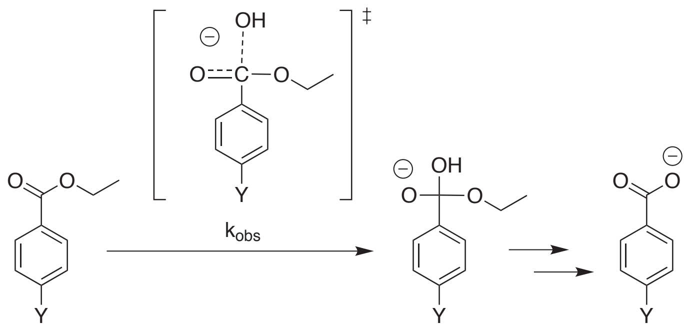


Fig. 17: Base hydrolysis of ethyl benzoate

Tab. 5: Some substituent parameters for the taft-Ingold equation.

Y	E_s	σ^*
—H	1.24	0.49
—Me	0	0
—Et	–0.07	–0.10
—CH_2Cl	–0.24	1.05
—$CHCl_2$	–1.54	1.94
—CCl_3	–2.06	2.65

Correlations using the Taft-Ingold equation for aliphatic substituents on reaction centers give adequate fits for many systems. The fits tend to be poorer than the Hammett system but, as you know, the Hammett system is "almost ideal" for linear correlations. The weakness of the Taft-Ingold system is that the assumption that there is no electronic influence in the acid hydrolysis is only approximately true. Thus, the substituent parameters still contain some information about electronic effects. Also the assumption that the steric effects will be the same for both base and acid hydrolysis of esters is also only approximately true. Thus the electronic reaction constants contain information about the steric effects of substituents. The Taft-Ingold parameters have not completely separated the two effects.

7 More and More. . .

In efforts to further improve these correlations many other parameter systems have been developed. Most attempt to use reference reaction systems where steric and electronic effects are more effectively isolated. For example, the electronic effects of substituents spaced away from the reaction center by rigid aliphatic rings could be better assumed to be "purely" electronic with no steric component. Electronic substituent constants established with these systems could then be used to establish steric constants in other systems.

We will not further pursue this area but there is much more to explore. Current research in this area involves using computational techniques to calculate the electronic perturbations at reaction center with various substituents. Of course, even these calculations must be calibrated to some reference reaction to have meaning in the real world.